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21CSE306P · APPLIED GENERATIVE AI · COURSE PROJECT 2026–27

VERITAS.

Evidence-Driven Generative AI for Reliable Research

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9 INDUSTRY, INNOVATION AND INFRASTRUCTURE

Problem statement

Generative AI can produce **inaccurate or unsupported information** with high confidence. Existing AI research tools often lack **claim-level verification** and **conflict detection**, making it difficult to assess the reliability of generated information.

Retrieval grounds a model in documents, but it can still add facts the evidence never states. **VERITAS catches hallucinations claim by claim.**

Objectives

- **O1** Ground answers in the user's documents
- **O2** Verify each claim against the evidence
- **O3** Detect conflicts between sources
- **O4** Give every claim a confidence score

End users

Student researchers

- Ask questions across their own papers and notes
- See which claims are verified, with source and page
- Cite only evidence-backed findings in their work

Data sources

User data — PDF, DOCX, TXT or MD uploaded by the user: the **only evidence** used

Demo corpus — 7 open-access arXiv papers on LLM hallucination

Benchmark — **SciFact** (Wadden et al., 2020): 5,183 abstracts, 300 expert-labelled claims

How VERITAS works: a fixed, single-pass generative pipeline

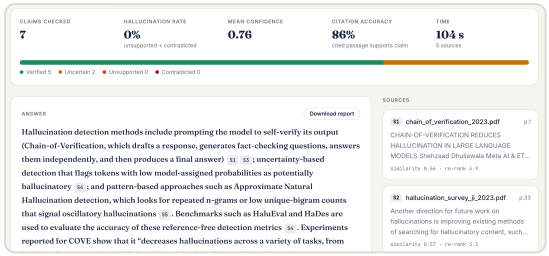
Knowledge indexing *once per document*	**Upload** User's documents, any topic	→ **Clean & chunk** ~220-word passages, bibliographies removed	→ **Embed** MiniLM: 384 numbers per passage	→ **Vector index** Search by cosine similarity
Grounded generation *RAG*	**Question** Any research question	→ **Retrieve** Top-20 → cross-encoder → best 5	→ **Grounded prompt** Only the evidence, cite every sentence	→ **Generate** gpt-oss-120b answer with [S#] citations
Verification & scoring *core contribution*	**Claim extraction** Answer split into atomic facts	→ **NLI check** DeBERTa-v3: entail, contradict or neither?	→ **Self-consistency** 4 re-samples: is the fact repeated?	→ **Verified report** Verified · Uncertain · Unsupported · Contradicted

confidence = 0.6 × NLI + 0.4 × consistency − 0.5 × contradiction

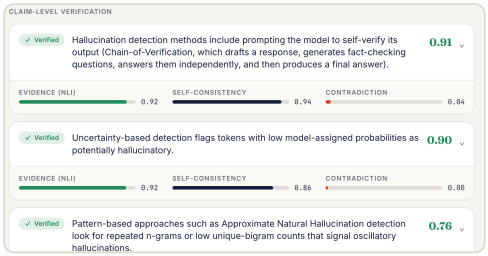
Verified ≥ 0.70 · Unsupported < 0.40 · corrected once · conflicts flagged

Software interface: VERITAS web app

live result on the 7 demo papers



Answer with [S#] citations, summary metrics and sources



Claim-level verification: NLI, self-consistency, contradiction

Prompt engineering

Role-play persona	**Instructional** grounding
Few-shot claim extraction	**Classification** judge
Chain-of-thought verdicts	**Comparison** conflicts
Information baseline	**Creative** summary

Models used

GENERATOR · gpt-oss-120b
open-weight LLM: writes answers, extracts claims

VERIFIER · DeBERTa-v3-base NLI
trained on MNLI, FEVER & ANLI: checks claims

EMBEDDER · all-MiniLM-L6-v2
384-dimensional embeddings for search

RE-RANKER · ms-marco-MiniLM-L-6-v2
cross-encoder: selects the 5 best passages

Future work

- Fine-tune the verifier on SciFact
- Learn the scoring weights from data
- Read tables and figures inside PDFs
- Complete the full RQ1, RQ3 and RQ5 study

Outputs & results

Generated output

"How can hallucinations in LLMs be detected?" (7 research papers)

- ✓ Token-level classifiers trained on synthetic noisy data can label each token as hallucinated or not. — **0.99**
- ✓ Hallucinatory outputs show concentrated attention and lower attention entropy. — **0.96**
- ✓ "Generate-then-refine" pipelines can identify and correct hallucinations. — **0.92**
- ✓ Factored verification variants improve performance. *NLI 1.00, but only 31% of re-samples repeat it* — **0.72**

Is the checker reliable?

SciFact, 340 pairs

0.70 accuracy
0.69 macro-F1
0.86 AUROC

	Supp.	Contra.	NEI
Supported	95	12	31
Contradicted	12	42	17
NEI	20	10	101

rows = experts · cols = model · NEI = not enough info

Prompt design vs. hallucination

Bad prompt (vague)	74%
Zero-shot instructions	8%
Role + instructions	0%
Few-shot	17%
Chain-of-thought	0%

74% → 0% same evidence, only the prompt changed (3-question preview)

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References: Lewis et al., RAG, NeurIPS 2020 · Wadden et al., SciFact, EMNLP 2020 · Manakul et al., SelfCheckGPT, EMNLP 2023 · Laban et al., SummaC, TACL 2022 · He et al., DeBERTa, ICLR 2021 · Vaswani et al., Attention Is All You Need, NeurIPS 2017